

Date of Birth
01/19/1994

Sex
Female

Physician
Anna Johnson

Institution
University of Kentucky
Multidisciplinary Clinic
110710532

TEMPUS | xT
648 gene panel

Tumor specimen:
Colon, sigmoid
UK Healthcare Enterprise Hospital
Laboratories - Pathology S24-03314, A1
Collected 2/2/2024
Received 4/2/2024
Tumor Percentage: 40%
(post microdissection)

Normal specimen:
Blood
Collected 3/28/2024
Received 3/30/2024

GENOMIC VARIANTS

Somatic - Potentially Actionable		Variant Allele Fraction
ATM	p.E28* Stop gain - LOF	5.4%
Somatic - Biologically Relevant		
SMAD4	p.G352R Missense variant - LOF	15.9%
TP53	p.V143A Missense variant - LOF	12.6%
KRAS	Copy number gain	

Tumor / Normal Matched Analysis (Potential Germline)

No potential germline variants were found in the limited set of genes on which we report.

This section reports variants in the tumor also seen in the normal. Only variants classified as pathogenic / likely pathogenic / risk allele are reported as potential germline. The normal match component has not been validated for hereditary cancer risk assessment. A validated test should be considered for hereditary risk assessment where indicated.

Pertinent Negatives

No pathogenic single nucleotide variants, indels, or copy number changes found in:

BRAF NRAS

IMMUNOTHERAPY MARKERS

Tumor Mutational Burden	Microsatellite Instability Status
1.1 m/MB 11th percentile	Stable Equivocal High

FDA-APPROVED THERAPIES, OTHER INDICATIONS

PARP Inhibitor	Olaparib	NCCN, Consensus, Prostate Cancer ATM p.E28* Loss-of-function
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None of the therapies on this report were identified in the clinical notes received and abstracted by Tempus, which may not reflect the complete treatment history.

Tumor-Agnostic Precision Immuno-Oncology and Somatic Targeting Rational for You (TAPISTRY) Platform Study (NCT04589845)	Phase II Cincinnati, OH - 75 mi ✓ ATM p.E28* mutation
Tovorafenib (DAY101) Monotherapy or in Combination With Other Therapies for Patients With Melanoma and Other Solid Tumors (NCT04985604)	Phase I/II Indianapolis, IN - 154 mi ✓ KRAS amplification
Study Of ATRN-119 In Patients With Advanced Solid Tumors (NCT04905914)	Phase I/II Cleveland, OH - 285 mi ✓ ATM p.E28* mutation ✓ TP53 p.V143A mutation

HUMAN LEUKOCYTE ANTIGEN (HLA) CLASS I GENOTYPING

HLA class I genotyping is determined by using DNA sequencing data and is only intended for clinical trial matching.

Gene	Allele I	Allele II
HLA-A	A*02:06^	A*24:02^
HLA-B	B*39:05^	B*40:02^
HLA-C	C*03:06	C*07:02^

^ This is the most representative allele within a group of alleles that are identical at exons 2 and 3 and cannot be disambiguated. For a full list of group members see Tempus Document Library.

SOMATIC VARIANTS OF UNKNOWN SIGNIFICANCE

No reportable variants of unknown significance (VUSs) were found.

LOW COVERAGE REGIONS

KDM5D

SOMATIC VARIANT DETAILS - POTENTIALLY ACTIONABLE

⬅️ ATM

c.82G>T p.E28* NM_000051 Stop gain - LOF

VAF: 5.4%

ATM is a member of the PI3/PI4-kinase gene family. ATM functions as a cell cycle checkpoint kinase, controlling the rate of cell growth and division. Additionally, it assists in the repair of damaged DNA. Loss of function mutations, copy number loss, and underexpression of ATM are associated with cancer progression.

↻

SMAD4

c.1054G>A p.G352R NM_005359 Missense variant - LOF

VAF: 15.9%

SMAD4 is a signal transduction protein that acts as the central mediator in the TGF-beta family signaling pathway through its interaction with upstream receptors and other SMAD transcription factors. The TGF-beta pathway is known to have a complex role in cancer development, progression, and metastasis. Loss of function mutations, copy number loss, and underexpression of SMAD4 are associated with cancer progression.

↻

TP53

c.428T>C p.V143A NM_000546 Missense variant - LOF

VAF: 12.6%

TP53 encodes a tumor suppressor that is commonly disabled across cancer types. It normally functions to activate cellular DNA repair mechanisms, plays a role in cell cycle progression in response to DNA damage, and can initiate apoptosis. Loss of function mutations, copy number loss, and epigenetic modifications resulting in underexpression of TP53 are associated with cancer progression.

⊖

KRAS

Copy number gain

KRAS is a GDP/GTP binding protein that acts as an intracellular signal transducer. KRAS is involved in several pathways involved in cellular proliferation and survival, including the PI3K-AKT-mTOR pathway and the Ras-Raf-MEK-ERK pathway. Activating mutations, copy number gains, and overexpression of KRAS are associated with cancer progression.

CLINICAL HISTORY

DIAGNOSIS

Diagnosed on
01/22/2024

THERAPIES

Oxaliplatin
Began 02/26/2024 - Ended unknown

Fluorouracil
Began 02/26/2024 - Ended unknown

Assay Description

The Tempus xT (version 4) assay is a custom oncology testing panel consisting of 648 genes with single nucleotide variants (SNVs), insertions and deletions (indels), copy number variants (CNVs), and chromosomal rearrangements (translocations) detected by hybrid capture next-generation sequencing (NGS) using custom designed IDT probes. A complete gene list can be found at the end of this assay description. This assay has 97.6% sensitivity for SNVs above 5% variant allele fraction (VAF), 91.4% sensitivity for indels above 5% VAF, and 96.3% sensitivity for reported chromosomal rearrangements. Certain driver or resistance genes may be reported to lower VAF when technically possible. The assay has 92.7% sensitivity for CNVs given the tumor purity limit of detection (LOD) for copy number gains at 30% and copy number loss at 40% tumor purity. The assay has 96.2% sensitivity for microsatellite instability (MSI) detection at 30% tumor purity and 97.6% sensitivity for tumor mutational burden high (TMB-high). Human leukocyte antigen (HLA) class I genotyping is reported from normal-matched analysis with 100% accuracy for HLA-A, 100% accuracy for HLA-B, and 100% accuracy for HLA-C; HLA genotyping from tumor analysis is reported with 100% accuracy for HLA-A, 97.8% accuracy for HLA-B, and 100% accuracy for HLA-C. Test results ordered before March 5, 2024 may not include HLA genotyping results.

Assay Description (continued)

EBF1, ECT2L, EGF, EGFR, EGLN1, EIF1AX, ELF3, ELOC (TCEB1), EMSY, ENG, EP300, EPCAM, EPHA2, EPHA7, EPHB1, EPHB2, EPOR, ERBB2 (HER2), ERBB3, ERBB4, ERCC1, ERCC2, ERCC3, ERCC4, ERCC5, ERCC6, ERG, ERFF1, ESR1, ETS1, ETS2, ETV1, ETV4, ETV5, ETV6, EWSR1, EZH2, FAM46C, FANCA, FANCB, FANCC, FANCD2, FANCE, FANCF, FANCG, FANCI, FANCL, FANCM, FAS, FAT1, FBXO11, FBXW7, FCGR2A, FCGR3A, FDP5, FGF1, FGF10, FGF14, FGF2, FGF23, FGF3, FGF4, FGF5, FGF6, FGF7, FGF8, FGF9, FGFR1, FGFR2, FGFR3, FGFR4, FH, FHIT, FLCN, FLT1, FLT3, FLT4, FNTB, FOXA1, FOXL2, FOXO1, FOXO3, FOXP1, FOXQ1, FRS2, FUBP1, FUS

G-H

G6PD, GABRA6, GALNT12, GATA1, GATA2, GATA3, GATA4, GATA6, GEN1, GLI1, GLI2, GNA11, GNA13, GNAQ, GNAS, GPC3, GPS2, GREM1, GRIN2A, GRM3, GSTP1, H19, H3F3A, HAS3, HAVCR2, HDAC1, HDAC2, HDAC4, HGF, HIF1A, HIST1H1E, HIST1H3B, HIST1H4E, HLA-A, HLA-B, HLA-C, HLA-DMA, HLA-DMB, HLA-DOA, HLA-DOB, HLA-DPA1, HLA-DPB1, HLA-DPB2, HLA-DQA1, HLA-DQA2, HLA-DQB1, HLA-DQB2, HLA-DRA, HLA-DRB1, HLA-DRB5, HLA-DRB6, HLA-E, HLA-F, HLA-G, HNF1A, HNF1B, HOXA11, HOXB13, HRAS, HSD11B2, HSD3B1, HSD3B2, HSP90AA1, HSPH1

I-K

IDH1, IDH2, IDO1, IFIT1, IFIT2, IFIT3, IFNAR1, IFNAR2, IFNGR1, IFNGR2, IFNL3, IKBKE, IKZF1, IL10RA, IL15, IL2RA, IL6R, IL7R, ING1, INPP4B, IRF1, IRF2, IRF4, IRS2, ITPKB, JAK1, JAK2, JAK3, JUN, KAT6A, KDM5A, KDM5C, KDM5D, KDM6A, KDR, KEAP1, KEL, KIF1B, KIT, KLF4, KLHL6, KLLN, KMT2A, KMT2B, KMT2C, KMT2D, KRAS

L-M

L2HGDH, LAG3, LATS1, LCK, LDLR, LEF1, LMNA, LMO1, LRP1B, LYN, LZTR1, MAD2L2, MAF, MAFB, MAGI2, MALT1, MAP2K1, MAP2K2, MAP2K4, MAP3K1, MAP3K7, MAPK1, MAX, MC1R, MCL1, MDM2, MDM4, MED12, MEF2B, MEN1, MET, MGMT, MIB1, MITF, MKI67, MLH1, MLH3, MLLT3, MN1, MPL, MRE11, MS4A1, MSH2, MSH3, MSH6, MTAP, MTHFD2, MTHFR, MTOR, MTRR, MUTYH, MYB, MYC, MYCL, MYCN, MYD88, MYH11

N-O

NBN, NCOR1, NCOR2, NF1, NF2, NFE2L2, NFKBIA, NHP2, NKX2-1, NOP10, NOTCH1, NOTCH2, NOTCH3, NOTCH4, NPM1, NQO1, NRAS, NRG1, NSD1, NSD2, NT5C2, NTHL1, NTRK1, NTRK2, NTRK3, NUDT15, NUP98, OLIG2

P-Q

P2RY8, PAK1, PALB2, PALLD, PAX3, PAX5, PAX7, PAX8, PBRM1, PCBP1, PDCD1, PDCD1LG2, PDGFRA, PDGFRB, PDK1, PHF6, PHGDH, PHLPP1, PHLPP2, PHOX2B, PIAS4, PIK3C2B, PIK3CA, PIK3CB, PIK3CD, PIK3CG, PIK3R1, PIK3R2, PIM1, PLCG1, PLCG2, PML, PMS1, PMS2, POLD1, POLE, POLH, POLQ, POT1, POU2F2, PPARA, PPARG, PPM1D, PPP1R15A, PPP2R2A, PPP6C, PRCC, PRDM1, PREX2, PRKAR1A, PRKDC, PRKN, PRSS1, PTCH1, PTCH2, PTEN, PTPN11, PTPN13, PTPN22, PTPRD, PTPRT, QKI

R-S

RAC1, RAD21, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RAD54L, RAF1, RANBP2, RARA, RASA1, RB1, RBM10, RECQL4, RET, RHEB, RHOA, RICTOR, RINT1, RIT1, RNF139, RNF43, ROS1, RPL5, RPS15, RPS6KB1, RPTOR, RRM1, RSF1, RUNX1, RUNX1T1, RXRA, SCG5, SDHA, SDHAF2, SDHB, SDHC, SDHD, SEC23B, SEMA3C, SETBP1, SETD2, SF3B1, SGK1, SH2B3, SHH, SLC26A3, SLC47A2, SLC9A3R1, SLIT2, SLX4, SMAD2, SMAD3, SMAD4, SMARCA1, SMARCA4, SMARCB1, SMARCE1, SMC1A, SMC3, SMO, SOCS1, SOD2, SOX10, SOX2, SOX9, SPEN, SPINK1, SPOP, SPRED1, SRC, SRSF2, STAG2, STAT3, STAT4, STAT5A, STAT5B, STAT6, STK11, SUFU, SUZ12, SYK, SYNE1

T-U

TAF1, TANC1, TAP1, TAP2, TARBP2, TBC1D12, TBL1XR1, TBX3, TCF3, TCF7L2, TCL1A, TERT, TET2, TFE3, TFEB, TFEC, TGFB1, TGFB2, TIGIT, TMEM127, TMEM173, TMPRSS2, TNF, TNFAIP3, TNFRSF14, TNFRSF17, TNFRSF9, TOP1, TOP2A, TP53, TP63, TPM1, TPMT, TRAF3, TRAF7, TSC1, TSC2, TSHR, TUSC3, TYMS, U2AF1, UBE2T, UGT1A1, UGT1A9, UMPS

V-Z

VEGFA, VEGFB, VHL, VSIR, WEE1, WNK1, WNK2, WRN, WT1, XPA, XPC, XPO1, XRCC1, XRCC2, XRCC3, YEATS4, ZFH3, ZMYM3, ZNF217, ZNF471, ZNF620, ZNF750, ZNRF3, ZRSR2

Gene Rearrangements Found by Targeted DNA Sequencing

ABL1, ALK, BCR, BRAF, EGFR, ETV6, EWSR1, FGFR2, FGFR3, MYB, NRG1, NTRK1, NTRK2, NTRK3, PAX8, PDGFRA, PML, RARA, RET, ROS1, TFE3, TMPRSS2

Hematologic Neoplasm Gene Rearrangements Found by Targeted DNA Sequencing

BCR-ABL1, PML-RARA

Potential Germline Finding Gene List

Assay Description (continued)

APC, ATM, AXIN2, BAP1, BARD1, BLM, BMPR1A, BRCA1, BRCA2, BRIP1, CDH1, CDK4, CDKN2A, CEBPA, CHEK2, DICER1, EGFR, EPCAM, ETV6, FH, FLCN, GATA2, KIT, MAX, MEN1, MET, MLH1, MSH2, MSH3, MSH6, MUTYH, NBN, NF1, NF2, NTHL1, PALB2, PDGFRA, PHOX2B, PMS2, POLD1, POLE, PRKAR1A, PTCH1, PTEN, RAD51C, RAD51D, RB1, RET, RUNX1, SDHA, SDHAF2, SDHB, SDHC, SDHD, SMAD4, SMARCA4, SMARCB1, STK11, SUFU, TMEM127, TP53, TSC1, TSC2, VHL, WT1

Tempus Disclaimer

The analysis of nucleic acids by next-generation sequencing (NGS) can be affected by multiple factors including formalin-fixation degrading DNA quality, and low tumor purity limiting sensitivity. Additionally, the chance of detecting genetic alterations may be reduced in regions of the genome which are structurally difficult to sequence, in homologous genes, or due to low complexity regions prone to sequencing error.

Genetic alterations are defined as clinically significant based on peer-reviewed published literature and other authoritative sources such as NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). These references are not comprehensive, therefore clinically unknown findings may occur.

These test results and information contained within the report are current as of the report date. Tempus will not update reports or send notification regarding reclassification of genomic alterations. This test was developed and its performance characteristics determined by Tempus AI, Inc. It has not been cleared or approved by the US Food and Drug Administration. The laboratory is CLIA certified to perform high-complexity testing. Any decisions related to patient care and treatment choices should be based on the independent judgment of the treating physician and should take into account all information related to the patient, including, without limitation the patient and family history, direct physical examination; and other tests. Tempus is not liable for medical judgment with regards to diagnosis, prognosis; or treatment in connection with the test results. Microdissection may have been performed at Tempus AI, Inc., either at 25 Alexandria Way, Ste 300, Durham, NC 27703 (CLIA# 34D2228314), or 600 West Chicago Ave, Ste 510, Chicago, IL 60654 (CLIA# 14D2114007). Bioinformatics processing was referred to and performed at Tempus AI, Inc. (CLIA# 14D2114007), 600 West Chicago Ave, Ste 510, Chicago, IL 60654.

Tempus may report certain potential secondary findings as part of these test results when a normal sample is provided, including potential germline and CHIP findings. Potential CHIP variants are intended only to inform the treating physician whether clonal hematopoiesis may impact the overall molecular profile of the tumor. The clinical significance and potential implications of potential CHIP findings are still under active investigation. The absence of reported CHIP results does not indicate the absence of potential clonal hematopoiesis nor of germline variants in the normal matched sample. The reportable potential germline secondary findings for the genes included in this panel include genes recommended for testing by the ACMG [1], the NCCN, and other published literature and are associated with inherited cancer syndromes. These secondary findings may or may not be related to the patient's current cancer diagnosis. The clinical significance of any such reported variants are based on germline classification criteria created by the ACMG/AMP [2]. Variants that are classified as pathogenic, likely pathogenic, or as a risk allele may be reported. Variants of uncertain significance (VUS), likely benign, and benign variants are not reported. Classifications are provided based on evidence evaluated at the time of reporting. When a variant is detected in both the somatic and germline samples, the variant is reported only under the "Tumor / Normal Matched Analysis (Potential Germline)" section of genomic variants, unless otherwise noted. Tempus does not notify physicians or patients of updated variant classifications.

This is not a stand alone germline test and has not been validated for assessment of hereditary disease risk. As such, the rate of false positives and false negatives has not been assessed and certain alterations are not able to be detected (e.g., single- and multi-exon level deletions/duplications and gene rearrangements). Additionally, detection of genetic variation in genes with high homology to other regions of the genome may be decreased or not reliably detected by NGS (including but not limited to these genes: NF1, PMS2, SBDS, and SUZ12). Mosaic germline alterations are not detectable by this assay. Because of these limitations, these germline test results cannot be used to definitively rule out cancer or other genetic predisposition syndromes. Unless Tempus has provided a separate report indicating that a specific germline finding is validated, the potential germline finding results set forth herein should not be used as a substitute for tests validated to determine genetic risk; additional validated hereditary testing may be recommended for potential germline findings not validated in a separate report from Tempus.

Results of genetic testing, including the potential germline findings described above, may have implications for both the patient and family members. Tempus does not provide genetic counseling; however, genetic counseling is strongly suggested, based on the patient's clinical history and/or genetic test results. The ordering physician or the patient is responsible for contacting a genetic counselor to discuss test results.

The clinical significance of specific HLA genotypes in drug response and prognosis is an area of active investigation. HLA class I allele typing results derived from this assay are intended for use in oncology clinical trials matching only, and are not validated as a means of genotyping for transplant workup. Findings are not prescriptive or conclusive for use of any specific therapeutic product. In some cases, a normal sample analysis may indicate a different call than what was provided from a previously reported tumor analysis; this gene-level reporting change will be indicated on the Amendment Report. Due to the limitations associated with short read technology, limited intronic coverage, and targeted analysis of exons 2 and 3, this test cannot disambiguate crossover events and a subset of ambiguous alleles cannot be excluded. Detection of one or more of these ambiguous HLA alleles will be designated on the report with a ^ icon; the full list of ambiguous alleles may be found in the Tempus Document Library. For tumor samples with high tumor purity, loss-of-heterozygosity may impact HLA genotype results called from the tumor instead of the normal. HLA genotyping results are not reported if Tempus's consent form signature requirements have not been met.

Tempus Disclaimer (continued)

For therapy entries designating that genes are wild-type, no activating variants were detected in the specified genes. Dostarlimab is indicated for the treatment of patients with mismatch repair deficient (dMMR) endometrial cancer or solid tumors. Although patients with MSI-H cancers are likely dMMR, confirmatory testing is recommended.

Dates and times are represented in the coordinated universal time zone (UTC) unless otherwise specified. However, dates that are provided to Tempus without a timestamp (e.g., sample collection date) are listed as provided.

1. Miller DT, Lee K, Chung WK, Gordon AS, Herman GE, Klein TE, Stewart DR, Amendola LM, Adelman K, Bale SJ, Gollob MH, Harrison SM, Hershberger RE, McKelvey K, Richards CS, Vlangos CN, Watson MS, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.0 list for reporting of secondary findings in clinical exome and genome sequencing: a policy statement of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2021 Aug;23(8):1381-1390. doi: 10.1038/s41436-021-01172-3. Epub 2021 May 20. Erratum in: Genet Med. 2021 Aug 3;; PMID: 34012068.

2. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehms HL; Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. ACMG Laboratory Quality Assurance Committee. Genet Med. 2015 May;17(5):405-24. DOI: 10.1038/gim.2015.30.

Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) Acute Lymphoblastic Leukemia Version: 2.2023, Acute Myeloid Leukemia Version: 6.2023, Ampullary Adenocarcinoma Version: 2.2023, Anal Carcinoma Version: 1.2024, Basal Cell Skin Cancer Version: 2.2024, B-Cell Lymphomas Version: 6.2023, Biliary Tract Cancers Version: 3.2023, Bladder Cancer Version: 3.2023, Bone Cancer Version: 1.2024, Breast Cancer Version: 4.2023, Central Nervous System Cancers Version: 1.2023, Cervical Cancer Version: 1.2024, Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma Version: 1.2024, Chronic Myeloid Leukemia Version: 1.2024, Colon Cancer Version: 4.2023, Dermatofibrosarcoma Protuberans Version: 1.2024, Esophageal and Esophagogastric Junction Cancers Version: 3.2023, Gastric Cancer Version: 2.2023, Gastrointestinal Stromal Tumors Version: 1.2023, Gestational Trophoblastic Neoplasia Version: 1.2024, Hairy Cell Leukemia Version: 1.2024, Head and Neck Cancers Version: 1.2024, Hepatobiliary Cancers Version: 1.2023, Hepatocellular Carcinoma Version: 2.2023, Histiocytic Neoplasms Version: 1.2023, Hodgkin Lymphoma Version: 1.2024, Kaposi Sarcoma Version: 1.2024, Kidney Cancer Version: 2.2024, Melanoma: Cutaneous Version: 3.2023, Melanoma: Uveal Version: 1.2023, Merkel Cell Carcinoma Version: 1.2023, Mesothelioma: Peritoneal Version: 1.2024, Mesothelioma: Pleural Version: 1.2024, Multiple Myeloma Version: 2.2024, Myelodysplastic Syndromes Version: 3.2023, Myeloid/Lymphoid Neoplasms with Eosinophilia and Tyrosine Kinase Gene Fusions Version: 1.2023, Myeloproliferative Neoplasms Version: 2.2023, Neuroendocrine and Adrenal Tumors Version: 1.2023, Non-Small Cell Lung Cancer Version: 5.2023, Occult Primary Version: 1.2024, Ovarian Cancer/Fallopian Tube Cancer/Primary Peritoneal Cancer Version: 2.2023, Pancreatic Adenocarcinoma Version: 2.2023, Pediatric Acute Lymphoblastic Leukemia Version: 3.2024, Pediatric Aggressive Mature B-Cell Lymphomas Version: 1.2023, Pediatric Central Nervous System Cancers Version: 2.2023, Pediatric Hodgkin Lymphoma Version: 2.2023, Penile Cancer Version: 1.2024, Primary Cutaneous Lymphomas Version: 1.2024, Prostate Cancer Version: 4.2023, Rectal Cancer Version: 6.2023, Small Bowel Adenocarcinoma Version: 3.2023, Small Cell Lung Cancer Version: 2.2024, Soft Tissue Sarcoma Version: 2.2023, Squamous Cell Skin Cancer Version: 1.2024, Systemic Light Chain Amyloidosis Version: 1.2024, Systemic Mastocytosis Version: 4.2023, T-Cell Lymphomas Version: 1.2023, Testicular Cancer Version: 1.2023, Thymomas and Thymic Carcinomas Version: 1.2024, Thyroid Carcinoma Version: 4.2023, Uterine Neoplasms Version: 1.2024, Vulvar Cancer Version: 2.2024, Waldenström Macroglobulinemia / Lymphoplasmacytic Lymphoma Version: 1.2024, Wilms Tumor (Nephroblastoma) Version: 1.2023. Version: 1.2023 © National Comprehensive Cancer Network, Inc. 2024. All rights reserved. Accessed November 26, 2023. To view the most recent and complete version of the guideline, go online to [NCCN.org](https://www.nccn.org).

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Date of Birth
01/19/1994

Sex
Female

Physician
Anna Johnson

Institution
University of Kentucky
Multidisciplinary Clinic
110710532

TEMPUS | xF
105 gene liquid biopsy

cfDNA specimen:
Peripheral Blood
Collected 3/28/2024
Received 3/30/2024

NO POTENTIALLY ACTIONABLE VARIANTS AND NO REPORTABLE
TREATMENT OPTIONS FOUND.

GENOMIC VARIANTS

Biologically Relevant	Variant Allele Fraction
 SMAD4 p.G352R Missense variant - LOF	71.9% 
 TP53 p.V143A Missense variant - LOF	68.8% 

Median Variant Allele Fraction

70.3%

IMMUNOTHERAPY MARKERS

Blood Tumor Mutational Burden (bTMB)	Microsatellite Instability Status
 8.6 m/MB	MSI-High not detected

The probability of having a tissue TMB >= 10 m/Mb if bTMB is >= 26 m/Mb is 63% and the probability of having a tissue TMB < 10 m/Mb if bTMB is < 26 m/Mb is 89%. The clinical performance of xF bTMB has not been determined. Tissue-based testing is recommended to determine the appropriateness of immunotherapy.

TREATMENT IMPLICATIONS

No reportable treatment options found.

CLINICAL TRIALS

Study Of ATRN-119 In Patients With Advanced Solid Tumors (NCT04905914)	Phase I/II Cleveland, OH - 285 mi ✓ TP53 p.V143A mutation
Immunotherapy Using Tumor Infiltrating Lymphocytes for Patients With Metastatic Cancer (NCT01174121)	Phase II Bethesda, MD - 406 mi ✓ TP53 p.V143A mutation

Trifluridine/Tipiracil and Talazoparib for the Treatment of Patients With Locally Advanced or Metastatic Colorectal or Gastroesophageal Cancer (NCT04511039)

Phase I
Buffalo, NY - 448 mi
✓ TP53 p.V143A mutation

VARIANTS OF UNKNOWN SIGNIFICANCE

Gene	Mutation effect	Variant allele fraction
NOTCH1	c.4571C>T p.A1524V Missense variant NM_017617	0.7%
EGFR	c.1627G>A p.E543K Missense variant NM_005228	0.4%
AKT1	c.1022A>G p.E341G Missense variant NM_005163	0.2%

LOW COVERAGE REGIONS

ERRFI1	JAK1	SPOP	TERT
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VARIANT DETAILS - BIOLOGICALLY RELEVANT

↻ SMAD4

c.1054G>A p.G352R NM_005359 Missense variant - LOF

VAF: 71.9%

SMAD4 is a signal transduction protein that acts as the central mediator in the TGF-beta family signaling pathway through its interaction with upstream receptors and other SMAD transcription factors. The TGF-beta pathway is known to have a complex role in cancer development, progression, and metastasis. Loss of function mutations, copy number loss, and underexpression of SMAD4 are associated with cancer progression.

↻ TP53

c.428T>C p.V143A NM_000546 Missense variant - LOF

VAF: 68.8%

TP53 encodes a tumor suppressor that is commonly disabled across cancer types. It normally functions to activate cellular DNA repair mechanisms, plays a role in cell cycle progression in response to DNA damage, and can initiate apoptosis. Loss of function mutations, copy number loss, and epigenetic modifications resulting in underexpression of TP53 are associated with cancer progression.

CLINICAL HISTORY

DIAGNOSIS

Diagnosed on
01/22/2024

THERAPIES

Oxaliplatin
Began 02/26/2024 - Ended unknown

THERAPIES

Fluorouracil
Began 02/26/2024 - Ended unknown

Assay Description

The Tempus xF assay is a next-generation sequencing (NGS) cell-free DNA liquid biopsy tumor profiling assay for identifying genomic alterations derived from solid tumors but circulating in the blood. The 105 gene panel includes single nucleotide variants (SNVs), insertions and deletions (indels), copy number variants (CNVs), and chromosomal rearrangements (translocations) detected by hybrid capture NGS using custom designed IDT probes. The assay typically uses 30 to 50 ng of input DNA, and at 30 ng of input material, the technical sensitivity is >99% for SNVs and CNV amplifications at ≥0.5% variant allele fraction (VAF), and >98% for indels and >97% for translocations at >0.5% VAF. The assay spans clinically relevant coding exons for 35 genes and covers recurrent hotspot mutations in 70 genes. Insertions and deletions will be reported down to the lower limit of detection (LLOD) in clinically relevant regions in 97 genes (list available upon request). The probability of having a tissue TMB ≥ 10 m/Mb if bTMB is ≥ 26 m/Mb is 63% and the probability of having a tissue TMB < 10 m/Mb if bTMB is < 26 m/Mb is 89%. The clinical performance of xF bTMB has not been determined. The assay may be run at 10 to <30 ng of input DNA, but in such a case, the report will indicate reduced sensitivity and consideration should be given to additional testing. Please see the [Tempus website](#) for a complete gene list and performance specifications.

Potentially Actionable alterations are protein-altering variants with an associated therapy, diagnostic, and/or prognostic indication, based on evidence from clinical guidelines and medical literature. **Biologically Relevant** variants have functional significance or an association with the disease state in the medical literature, but do not have relevant therapeutic, prognostic or diagnostic evidence in the Tempus knowledge database. **Variants of Unknown Significance (VUSs)** exhibit an unclear effect on function and/or do not have sufficient evidence to determine their pathogenicity. **Benign variants** are not reported. **Low Coverage Regions** are included when mean coverage over any region(s) of a gene falls below a threshold of 1000x. The absence of alterations in genes with low coverage should be interpreted carefully in the context of the patient's diagnosis with consideration for retesting. Variants are identified through aligning the patient's DNA sequence to the human genome reference sequence version hg19 (GRCh37). The clinical summary shows actionable and biologically relevant variants. Because sequencing is performed without a matched normal sample, it is not possible to distinguish whether reported variants are germline or somatic.

Blood tumor mutational burden (bTMB) measures the quantity of somatic SNVs and indels, of any pathogenicity, including benign, carried in a tumor as the number of protein-altering mutations per million coding base pairs. Variants used in this calculation may not necessarily appear on the xF report. The bTMB value for a patient may differ from the tissue TMB value measured from the same patient. There are various reasons why bTMB is generally higher than tissue TMB including tumor heterogeneity and difference in sensitivities and panel size of this liquid biopsy and solid tumor assays. Studies have shown that tumors with higher tissue TMB have an increased likelihood of response to immunotherapy [1, 2].

Microsatellite instability (MSI) refers to hypermutability caused by genetic or acquired defects in the DNA mismatch repair pathway. **MSI-high (MSI-H)** tumors have changes in microsatellite repeat lengths due to defective DNA mismatch repair activity. MSI-H status is reported when detected. If MSI status will affect clinical management, immunohistochemical staining for DNA mismatch repair proteins, or another method of ascertaining MSI status, is recommended.

1. Nivolumab plus Ipilimumab in Lung Cancer with a High Tumor Mutational Burden. <https://www.ncbi.nlm.nih.gov/pubmed/29658845>
- 2.Cancer immunology. Mutational landscape determines sensitivity to PD-1 blockade in non-small cell lung cancer. <https://www.ncbi.nlm.nih.gov/pubmed/25765070>

xF provides insights into the clinically relevant biomarkers incorporated in OncoKB, NCCN and other oncology guidelines for:

- Bladder cancer: FGFR2, FGFR3
- Breast cancer: BRCA1, BRCA2, ERBB2 (HER2), ESR1, PIK3CA
- Cholangiocarcinoma: FGFR2, IDH1
- Colorectal cancer: BRAF, ERBB2 (HER2), KRAS, NRAS
- Gastroesophageal adenocarcinoma: ERBB2 (HER2)
- Gastrointestinal stromal tumor: KIT, PDGFRA
- Melanoma: BRAF, KIT, NRAS
- Non-small cell lung cancer: ALK, BRAF, EGFR, ERBB2 (HER2), KRAS, MET, RET, ROS1

Complete Gene List

Assay Description (continued)**A-C**

AKT1, AKT2, ALK, APC, AR, ARAF, ARID1A, ATM, ATR, B2M, BAP1, BRAF, BRCA1, BRCA2, BTK, CCND1, CCND2, CCND3, CCNE1, CD274 (PD-L1), CDH1, CDK4, CDK6, CDKN2A, CTNNB1

D-F

DDR2, DPYD, EGFR, ERBB2 (HER2), ERFF1, ESR1, EZH2, FBXW7, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2

G-M

GATA3, GNA11, GNAQ, GNAS, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KEAP1, KIT, KMT2A, KRAS, MAP2K1, MAP2K2, MAPK1, MET, MLH1, MPL, MSH2, MSH3, MSH6, MTOR, MYC, MYCN

N-R

NF1, NF2, NFE2L2, NOTCH1, NPM1, NRAS, NTRK1, PALB2, PBRM1, PDCD1LG2, PDGFRA, PDGFRB, PIK3CA, PIK3R1, PMS2, PTCH1, PTEN, PTPN11, RAD51C, RAF1, RB1, RET, RHEB, RHOA, RIT1, RNF43, ROS1

S-Z

SDHA, SMAD4, SMO, SPOP, STK11, TERT, TP53, TSC1, TSC2, UGT1A1, VHL

Gene Rearrangements

ALK, BRAF, FGFR2, FGFR3, NTRK1, RET, ROS1

Copy Number Gains

CCNE1, CD274 (PD-L1), EGFR, ERBB2 (HER2), MET, MYC

Tempus Disclaimer

The analysis of nucleic acids by next-generation sequencing (NGS) can be affected by multiple factors including DNA quality, hemolysis of the peripheral blood sample, and low amounts of circulating cell-free DNA limiting sensitivity. Tempus may report findings below our sensitivity threshold due to reduced sample quality and/or quantity. For samples flagged as falling below this threshold, Tempus advises resequencing in order to provide more accurate results. Additionally, the chance of detecting genetic alterations may be reduced in regions of the genome which are structurally difficult to sequence, in homologous genes, or due to sequencing errors.

Genetic alterations are defined as clinically significant based on peer-reviewed published literature and other authoritative sources such as NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). These references are not comprehensive, therefore clinically unknown findings may occur.

These test results and information contained within the report are current as of the report date. Tempus will not update reports or send notification regarding reclassification of genomic alterations. This test was developed and its performance characteristics determined by Tempus AI, Inc. It has not been cleared or approved by the US Food and Drug Administration. The laboratory is CLIA certified to perform high-complexity testing. Any decisions related to patient care and treatment choices should be based on the independent judgment of the treating physician and should take into account all information related to the patient, including without limitation, the patient and family history, direct physical examination and other tests. Tempus is not liable for medical judgment with regards to diagnosis, prognosis or treatment in connection with the test results. Bioinformatics processing was referred to and performed at Tempus AI, Inc. (CLIA# 14D2114007), 600 West Chicago Ave, Ste 510, Chicago, IL 60654

Note that certain tumor type-, sample- or variant-related characteristics, such as low cell-free DNA concentration, may result in reduced analytic sensitivity of the xF test for detection of alterations in the covered genes, including the above mentioned guideline-recommended genes.

bTMB results on this report, when determined, are not intended to replace tissue or larger panel analyses, but may help identify patients who should be considered for TMB analysis using solid tissue. Follow-up testing using solid tissue is recommended to determine the appropriateness of immunotherapy. Any clinical decision, including a decision to obtain tissue for TMB testing, should be made in the context of the patient's entire medical history, not on bTMB results alone.

Different therapies indicated for the treatment of patients with mismatch repair deficient (dMMR) endometrial cancer or solid tumors. Although patients with MSI-H cancers are likely dMMR, confirmatory testing is recommended.

Dates and times are represented in the coordinated universal time zone (UTC) unless otherwise specified. However, dates that are provided to Tempus without a timestamp (e.g., sample collection date) are listed as provided.

This assay cannot distinguish whether cell-free variants detected in plasma are derived from a patient's solid tumor or from clonal blood cells. Variants associated with clonal hematopoietic processes may be detected in a high percentage of older individuals and are especially common in the following genes:

ATM, BRAF, BRCA1, BRCA2, EZH2, FLT3, GNAS, IDH1, IDH2, JAK2, KIT, KRAS, NOTCH1, NPM1, NRAS, PALB2, PTPN11, RAD51C, RHOA, TP53

Tempus Disclaimer (continued)

Correlation with tumor and/or blood sequencing results may be helpful in clarifying the origin of variants in some cases.

1. Genovese G, Kahler AK, Handsaker RE, et al. Clonal hematopoiesis and blood-cancer risk inferred from blood DNA sequence. N Engl J Med. 2014;371:2477-2487.
2. Jaiswal S, Fontanillas P, Flannick J, et al. Age-related clonal hematopoiesis associated with adverse outcomes. N Engl J Med. 2014;371:2488-2498.
3. Chan HT, Chin YM, Nakamura Y, et al. Clonal Hematopoiesis in Liquid Biopsy: From Biological Noise to Valuable Clinical Implications. Cancers 2020, 12, 2277. [Clonal Hematopoiesis in Liquid Biopsy: From Biological Noise to Valuable Clinical Implications](#)

Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) Acute Lymphoblastic Leukemia Version: 2.2023, Acute Myeloid Leukemia Version: 6.2023, Ampullary Adenocarcinoma Version: 2.2023, Anal Carcinoma Version: 1.2024, Basal Cell Skin Cancer Version: 2.2024, B-Cell Lymphomas Version: 6.2023, Biliary Tract Cancers Version: 3.2023, Bladder Cancer Version: 3.2023, Bone Cancer Version: 1.2024, Breast Cancer Version: 4.2023, Central Nervous System Cancers Version: 1.2023, Cervical Cancer Version: 1.2024, Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma Version: 1.2024, Chronic Myeloid Leukemia Version: 1.2024, Colon Cancer Version: 4.2023, Dermatofibrosarcoma Protuberans Version: 1.2024, Esophageal and Esophagogastric Junction Cancers Version: 3.2023, Gastric Cancer Version: 2.2023, Gastrointestinal Stromal Tumors Version: 1.2023, Gestational Trophoblastic Neoplasia Version: 1.2024, Hairy Cell Leukemia Version: 1.2024, Head and Neck Cancers Version: 1.2024, Hepatobiliary Cancers Version: 1.2023, Hepatocellular Carcinoma Version: 2.2023, Histiocytic Neoplasms Version: 1.2023, Hodgkin Lymphoma Version: 1.2024, Kaposi Sarcoma Version: 1.2024, Kidney Cancer Version: 2.2024, Melanoma: Cutaneous Version: 3.2023, Melanoma: Uveal Version: 1.2023, Merkel Cell Carcinoma Version: 1.2023, Mesothelioma: Peritoneal Version: 1.2024, Mesothelioma: Pleural Version: 1.2024, Multiple Myeloma Version: 2.2024, Myelodysplastic Syndromes Version: 3.2023, Myeloid/Lymphoid Neoplasms with Eosinophilia and Tyrosine Kinase Gene Fusions Version: 1.2023, Myeloproliferative Neoplasms Version: 2.2023, Neuroendocrine and Adrenal Tumors Version: 1.2023, Non-Small Cell Lung Cancer Version: 5.2023, Occult Primary Version: 1.2024, Ovarian Cancer/Fallopian Tube Cancer/Primary Peritoneal Cancer Version: 2.2023, Pancreatic Adenocarcinoma Version: 2.2023, Pediatric Acute Lymphoblastic Leukemia Version: 3.2024, Pediatric Aggressive Mature B-Cell Lymphomas Version: 1.2023, Pediatric Central Nervous System Cancers Version: 2.2023, Pediatric Hodgkin Lymphoma Version: 2.2023, Penile Cancer Version: 1.2024, Primary Cutaneous Lymphomas Version: 1.2024, Prostate Cancer Version: 4.2023, Rectal Cancer Version: 6.2023, Small Bowel Adenocarcinoma Version: 3.2023, Small Cell Lung Cancer Version: 2.2024, Soft Tissue Sarcoma Version: 2.2023, Squamous Cell Skin Cancer Version: 1.2024, Systemic Light Chain Amyloidosis Version: 1.2024, Systemic Mastocytosis Version: 4.2023, T-Cell Lymphomas Version: 1.2023, Testicular Cancer Version: 1.2023, Thymomas and Thymic Carcinomas Version: 1.2024, Thyroid Carcinoma Version: 4.2023, Uterine Neoplasms Version: 1.2024, Vulvar Cancer Version: 2.2024, Waldenström Macroglobulinemia / Lymphoplasmacytic Lymphoma Version: 1.2024, Wilms Tumor (Nephroblastoma) Version: 1.2023. Version: 1.2023 © National Comprehensive Cancer Network, Inc. 2024. All rights reserved. Accessed November 26, 2023. To view the most recent and complete version of the guideline, go online to NCCN.org. The NCCN Guidelines® and other content provided by NCCN are works in progress that may be refined as often as new significant data becomes available. They are statements of consensus of its authors regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult any NCCN Guidelines® or other NCCN Content is expected to use independent medical judgment in the context of individual clinical circumstances to determine any patient's care or treatment. NCCN makes no warranties of any kind whatsoever regarding their content, use or application and disclaims any responsibility for their application or use in any way. Therapeutic options are not applicable in all disease settings.

The OncoKB™ precision oncology knowledge base was made available under license from Memorial Sloan Kettering Cancer Center. See terms [here](#).

Date of Birth
01/19/1994

Sex
Female

Physician
Anna Johnson

Institution
**University of Kentucky
Multidisciplinary Clinic
110710532**

TEMPUS | IHC

Analysis performed using DAKO PD-L1
22C3 clone.

Tumor specimen:
Colon, sigmoid
UK Healthcare Enterprise Hospital
Laboratories - Pathology S24-03314, A1
Collected 2/2/2024
Received 4/2/2024

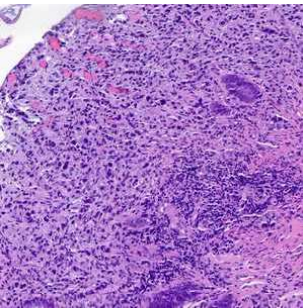
PD-L1 EXPRESSION

Tumor Proportion Score (TPS)

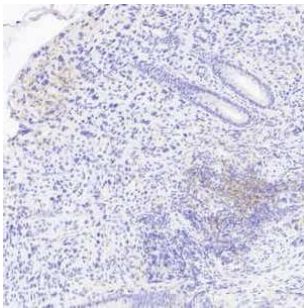
1%

Combined Positive Score (CPS)

5



H&E



PD-L1

FDA approved companion diagnostic interpretation cutoffs are based on tumor subtype. Approved indications are as follows:

Tumor Proportion Score (TPS) Expression Levels

Tumor indication	Expression level
Non-Small Cell Lung Cancer	≥ 1%
Non-Small Cell Lung Cancer (high positive)	≥ 50%

Combined Positive Score (CPS) Expression Levels

Tumor indication	Expression level
Cervical Cancer	≥ 1
Esophageal Squamous Cell Carcinoma	≥ 10
Head and Neck Squamous Cell Carcinoma	≥ 1
Triple-Negative Breast Cancer	≥ 10

Interpretation Guidelines

The Tempus PD-L1 22C3 IHC test is a qualitative immunohistochemical assay using monoclonal mouse anti-PD-L1, Clone 22C3 intended for use in the detection of PD-L1 protein in formalin-fixed, paraffin-embedded (FFPE) tissues using the Leica BOND system. The Tumor Proportion Score (TPS) is the percentage of viable tumor cells showing partial or complete membrane staining at any intensity. The Combined Positive Score (CPS) is the number of programmed death-ligand 1 (PD-L1) staining cells (tumor cells, lymphocytes, macrophages) divided by the total viable tumor cells, multiplied by 100. Although the result of the calculation can exceed 100, the maximum score is defined as CPS 100. Scoring interpretation is not provided in tumors for which no scoring system has been published. See clinical practice guidelines for specific clinical circumstances guiding PD-L1 testing. Results (positive/negative) are determined based on available clinical information provided at the time of test interpretation. Test results should be interpreted in the context of clinical findings by a treating physician.

Tempus Disclaimer

This test was developed and its performance characteristics determined by Tempus AI, Inc. It has not been cleared or approved by the US Food and Drug Administration. The laboratory is CLIA certified to perform high-complexity testing. All controls were reviewed and show appropriate positive or negative reactivity. This assay has not been validated on decalcified tissue or cytology specimens; results from these tissue types should be interpreted with caution. Cytology specimens may not contain stroma containing immune cells. As a result, the Combined Positive Score (CPS) for this patient's tumor may be underestimated. No Tumor Proportion Score (TPS) scores are provided for lymphoma, leukemia, or other hematological malignancy cases. In reports containing photographic images, those images should be considered representative and should not be used for diagnostic purposes. These test results and information contained within the report are current as of the report date. Any decisions related to patient care and treatment choices should be based on the independent judgment of the treating physician and should take into account all information related to the patient, including without limitation, the patient and family history, direct physical examination and other tests. Tempus is not liable for medical judgment with regards to diagnosis, prognosis or treatment in connection with the test results.

Dates and times are represented in the coordinated universal time zone (UTC) unless otherwise specified. However, dates that are provided to Tempus without a timestamp (e.g., sample collection date) are listed as provided.

Date of Birth
01/19/1994

Sex
Female

Physician
Anna Johnson

Institution
University of Kentucky
Multidisciplinary Clinic
110710532

TEMPUS | xR

Transcriptome

Tumor specimen:
Colon, sigmoid
UK Healthcare Enterprise Hospital
Laboratories - Pathology S24-03314, A1
Collected 2/2/2024
Received 4/2/2024
Tumor Percentage: 40%
(post microdissection)

Normal specimen:
Blood
Collected 3/28/2024
Received 3/30/2024

Notes

The reported chromosomal rearrangement has not been previously reported or functionally characterized. Please refer to the Variant Details section for more information.

THIS REPORT IS BEING ISSUED TO REPORT THE RESULTS OF GENE REARRANGEMENT AND ALTERED SPLICING ANALYSIS FROM RNA SEQUENCING.

RNA sequencing analysis identified a FGFR3 - DSP rearrangement.

GENOMIC VARIANTS

Somatic - Potentially Actionable

FGFR3-DSP Chromosomal rearrangement

FDA-APPROVED THERAPIES, OTHER INDICATIONS

Pan-FGFR Inhibitor	Erdafitinib	NCCN, Consensus, Bladder Cancer FGFR3-DSP Chromosomal rearrangement
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We were unable to determine whether treatments on this report were previously prescribed for this patient.

CLINICAL TRIALS

TAPUR: Testing the Use of Food and Drug Administration (FDA) Approved Drugs That Target a Specific Abnormality in a Tumor Gene in People With Advanced Stage Cancer (NCT02693535)	Phase II Cincinnati, OH - 75 mi ✓ FGFR3-DSP chromosomal rearrangement
A Study of LOXO-435 in Participants With Cancer With a Change in a Gene Called FGFR3 (NCT05614739)	Phase I Nashville, TN - 180 mi ✓ FGFR3-DSP chromosomal rearrangement
Safety and Preliminary Anti-Tumor Activity of TYRA-300 in Advanced Urothelial Carcinoma and Other Solid Tumors With FGFR3 Gene Alterations (NCT05544552)	Phase I/II Nashville, TN - 182 mi ✓ FGFR3-DSP chromosomal rearrangement

FGFR3-DSP Chromosomal rearrangement

FGFR3 encodes a receptor protein in the fibroblast growth factor family. These receptors are involved in multiple pathways associated with cellular proliferation and survival processes, including the Ras-Raf-MEK-ERK and the PI3K-AKT-mTOR pathways. The 5' portion of the FGFR3 gene containing the tyrosine kinase domain can fuse to the 3' portion of several other genes resulting in constitutively active FGFR3 tyrosine kinase activity.

Assay Description

The Tempus xR test is a whole-transcriptome genomic sequence analysis panel for solid tumor and hematological malignancy samples that reports clinically relevant chromosomal rearrangements arising from the genetic rearrangement of more than 100 targeted genes, as well as altered splicing for MET Exon 14 and EGFRvIII, in an unbiased and comprehensive manner. Chromosomal rearrangements and altered splicing events are detected by hybrid capture next generation sequencing (NGS) using custom designed IDT probes. Test results are intended to provide tumor molecular information that may be used by clinicians to help inform clinical management when patients are seeking further cancer treatment. This assay has 96.8% sensitivity for targeted chromosomal rearrangements, 100% sensitivity for untargeted chromosomal rearrangements, and 100% sensitivity for altered splicing at 20% tumor purity.

The clinical summary (first page of the report) shows potentially actionable and biologically relevant chromosomal rearrangements and altered splicing variants. RNA chromosomal rearrangement and altered splicing event analysis is performed without a matched normal sample and so results do not distinguish whether reported chromosomal rearrangements or altered splicing events are germline or somatic. **Potentially Actionable** alterations have an associated therapeutic, diagnostic, and/or prognostic indication, based on evidence from US-based clinical guidelines and medical literature curated in the Tempus knowledge database. **Biologically Relevant** alterations may have functional significance or have been observed in the medical literature, but do not have relevant therapeutic, diagnostic, or prognostic evidence curated in the Tempus knowledge database. Benign variants are not reported. Alterations are identified by aligning the patient's RNA sequence to the human genome reference sequence version hg19 (GRCh37).

The xR assay uses IDT xGen Exome Research Panel v2 probe set as backbone, which consists of >415K individually synthesized probes and spans 34 Mb target region (19,433 genes) of the human genome. Additional Tempus-specific custom spike-in probes are added to enhance target region detection. When whole transcriptome RNA-Seq is performed, expressed fusion transcripts from chromosomal rearrangements specifically targeted by the assay will be detected. In addition, expressed fusion transcripts from rearranged genes not specifically targeted by the assay may also be detected. A list of targeted fusion genes and targeted altered splicing events can be made available upon request. The bioinformatics chromosomal rearrangement transcript detection pipeline identifies and analyzes the positions of breakpoint-spanning reads and split paired-end reads. Non-canonical chromosomal rearrangement transcripts may be reported at the discretion of the pathologist. The altered splicing bioinformatics pipeline analyzes the positions of split paired-end reads spanning annotated and novel exon-exon junctions.

Tempus Disclaimer

The analysis of nucleic acids by next-generation sequencing (NGS) can be affected by multiple factors including formalin-fixation, degrading RNA quality, and low tumor purity limiting sensitivity. Additionally, the chance of detecting genetic alterations may be reduced in regions of the genome which are structurally difficult to sequence, in homologous genes, or due to low complexity regions that are prone to sequencing error.

Genetic alterations are defined as clinically significant based on peer-reviewed published literature and other authoritative sources such as NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). These references are not comprehensive, therefore clinically unknown findings may occur. Tempus does not notify physicians or patients of updated variant classifications.

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Tempus Disclaimer (continued)

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